🐱 ARDUINO FIRMWARE · VERSION V12 · 2026

J-DRUMS

The electronic drum MIDI module built from scratch with Arduino Mega 2560 — powerful, configurable, and open-source.

Arduino Mega 2560 | MIT License | 16 Analog Pads | MIDI USB + Serial | Smart Hi-Hat

ABOUT THE PROJECT

A complete and customizable MIDI module

> *J-DRUMS V12 is an electronic drum MIDI controller built for the Arduino Mega 2560. It turns piezos and sensors into professional MIDI signals — with an on-screen LCD menu, individual per-pad adjustment, a smart hi-hat via TCRT5000, and compatibility with all the leading VST software on the market.*

KEY FEATURES

Everything you need in the kit

🎛

16 Analog Pads (A0–A15)

Each pad has an individually configurable MIDI note, threshold, gain, velocity curve, and crosstalk cancellation.

🎚

Smart Hi-Hat (TCRT5000)

Infrared optical sensor controls hi-hat opening in real time — auto-calibration in 8 seconds with A+B. Supports **two modes via InvSensor**: *Normal* (sensor manually assembled with a resistor) and *Inverted* (ready-made module with LM393 — e.g. TCRT5000 Arduino LM393 5V sensor). Calibration works the same in both modes.

🎹

3 Factory Presets with Ready-Made MIDI Maps

Addictive, SupEzd1, SupEzd2 — mapping switches automatically when you select the preset.

📡

Dual MIDI Output

USB and Serial TX1 work simultaneously — connect to your PC and a sound module at the same time.

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Digital Pad Expansion

7 built-in digital pads (pins D33–D45) + expansion via Pro Micro (4 pads) or Leonardo (6 pads) over SysEx.

💾

3 Preset Slots in EEPROM

Save and restore complete kit configurations — notes, thresholds, chokes, and everything else with one click.

🎯

XCancel CrossTalk

Eliminates ghost notes from mechanical vibration — configurable Source→Target per pad without losing dynamics.

⚡

Quick Pad Adjust

Hold the encoder for 3s, hit the pad, and the menu jumps straight to that pad's settings — no menu navigation needed.

🔊

Real-Time Monitor

MIDI Monitor shows port, note, and velocity for each hit on the LCD. Hi-Hat Monitor displays a 16-position visual bar.

🥁

Rimshot, Tri-Zone, and Dual Pad — Hi-Hat, Ride, and Crash with 1 piezo each

A complete multi-zone system using digital membranes (ZoneDual/Tri-Zone). A single piezo per cymbal detects body, edge, and bell separately.

🎚 HI-HAT — TRI-ZONE

Piezo on **A1** + 2 digital membranes:
● Edge/Rim → free piezo (note 8)
● Body → Dig 3 / P41 (note 7)
● Bell → Dig 4 / P39 (note 9)

Pedal opening via HHC (A0) works correctly across all *3 zones*.

♪ RIDE — TRI-ZONE

Piezo on **A8** + 2 digital membranes:
● Bow → free piezo (note 51)
● Bell → Dig 1 / P45 (note 53)
● Ride Crash → Dig 2 / P43 (note 59)

Real piezo velocity is inherited across all 3 zones.

💥 **CRASH — DUAL ZONE (×2)**

Crash 1 (A10) + Dig 5 / P37:
● Edge → free piezo (note 49)
● Body → membrane (note 27)

Crash 2 (A11) + Dig 6 / P35:
● Edge → free piezo (note 57)
● Body → membrane (note 31)

🥁 **RIMSHOT — 2 MODES**

Full rimshot:
Rim (A3) + Snare (A2) together → automatically triggers the rimshot note (40).

Force rimshot:
Snare (A2) at velocity 127 → automatically triggers rimshot, no rim hit needed.

Each mode is enabled independently in the menu.

⚠ **RESERVED PORTS — A0, A1, A2, A3, A8, A10, A11**

These ports have notes reserved by the system and **should not be changed by the user**:
A0 = HHC — Hi-Hat Controller (note managed by the Hi-Hat system)
A1 = Hi-Hat (note managed by the Hi-Hat system)
A2 = Snare (note managed by the Rimshot system)
A3 = Snare Rim (note managed by the Rimshot system)
A8 = Ride Tip (note managed by the Ride's Tri-Zone system)
A10 = CRASH 1 (note managed by Crash 1's Dual-Zone Body system)
A11 = CRASH 2 (note managed by Crash 2's Dual-Zone Body system)

Changing these notes can break hi-hat and rimshot operation.

⚡ **80ms time latch** — even if the stick touches the membrane for just 3–10ms, the system keeps the zone active long enough for the piezo to process the hit. Works with any homemade membrane: copper tape, EVA foam, sheet metal.

🎵

Cymbal Choke

4 chokes on pins D47–D53 — triggered by direct metal contact, no extra components needed.

🔀

InvSensor — Hi-Hat Sensor Mode

Supports two sensor types for the Hi-Hat Controller (A0): **Normal** — manually assembled sensor with transistor and resistor (TCRT5000 + R10kΩ); **Inverted** — ready-made module with integrated LM393 (e.g. TCRT5000 Arduino LM393 5V Infrared Sensor). Auto-calibration works the same in both modes. Configured under: Pad A0 → InvSensor. Saved to EEPROM.

🔄

Custom Velocity Curves

7 curves available per pad: Linear, Soft1/2/3, Hard1/2, and Fixed — adapt the response to your playing style.

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Fi-TipBell — Ride Tip×Bell Dispute Filter

An alternative to membrane Tri-Zone for the Ride: two separate piezos — tip on A8, bell on a second small cymbal on A9, isolated by just a nut between them. When both trigger within 12ms of each other, the firmware compares hit energy (velocity

÷ Gain) and sends only the note from whichever piezo was actually hit, with an adjustable bonus (0–30) to compensate for the bell's mechanically weaker signal. Only active on A8, and only while the Ride's Tri-Zone is Off.

🛡

Noise vs Real Hit Filter

A piezo never reads total silence — it picks up vibration from fans, TVs, switching power supplies, and the mains itself (60Hz). J-DRUMS detects the sharp variation of a real hit and ignores slow oscillations, blocking false triggers without touching real dynamics.

V12

Current firmware version — Arduino Mega 2560

16	1	4	3	4	16
ANALOG PADS	DIGITAL PAD	CHOKES	PRESET SLOTS	MIDI MAPS	XCANCEL PAIRS

54+

Tested and approved by a group of over 54 members, including professional musicians, drummers, and percussionists with years of experience, who test J-DRUMS on acoustic kits converted to electronic and use it in real practice.

> *The code reached such a high level that many entry-level modules on the market, with more hardware resources, deliver less performance and fewer features than J-DRUMS. Even running on an Arduino Mega 2560, the result achieved gets close to the level of top-tier professional modules, such as the TD-27 and similar.*

MODULE NAVIGATION

Full menu
on the LCD display

- 🥁 PADs A0–A15
- 🎚 Hi-Hat Controller
- ⚡ Quick Pad Adjust
- ⚙ Settings
- 💡 Display Backlight
- 🎯 Dual Pad
- 💾 Save Presets
- 🔊 Menu Sound
- 🎤 MIDI Output
- 🔌 Auxiliary Pads
- 🔄 Invert Buttons
- 📊 MIDI Monitor
- 👁 Hi-Hat Monitor
- 🔀 InvSensor (A0)
- 🎛 XCancel CrossTalk
- 🔄 Reset Module
- 🎼 Trizone Notes
- 🔇 Choke Notes

⚙ SETTINGS

All submenus
inside Settings

📋 Summary of every submenu available inside Settings — in firmware order.

💡 Display Backlight

Turns off the LCD after 30s of inactivity, or keeps it always on

🎯 Dual Pad

Enables/disables dual notes from a single piezo (101→102, 103→104)

Save Presets

Saves or restores the configuration in one of 3 EEPROM slots. Also includes 3 fixed factory presets (Addctive, SupEzd1, SupEzd2) that can never be erased.

Menu Sound

Enables or disables the audio feedback beeps during navigation

MIDI Output

Enables MIDI output via USB and/or TX1 — both can be used simultaneously

Auxiliary Pads

Enables pad expansion via Pro Micro (4 pads) or Leonardo (6 pads) over SysEx

Invert Buttons

Swaps the function of Buttons A and B between pins 6 and 7 — saved to EEPROM

Trizone Notes

Sets the MIDI note for the 7 digital pads on pins D45,43,41,39,37,35,33

Choke Notes

Sets the MIDI note for the 4 cymbal chokes on pins D53,51,47,49

Reset Module

Restores every value to factory default — double confirmation with a 10-second timeout each

MIDI Monitor

Monitors in real time the analog port (A0–A15), the MIDI note, and the velocity of every hit — MIDI keeps working normally during monitoring

Hi-Hat Monitor

Monitors hi-hat opening via the TCRT5000 with a 16-position visual bar — 1 block = maximum opening, 16 blocks = fully closed. Line 0: A<118> Hi F<001> with calibration adjustable from the menu.

CC MIDI Preset

Saves or restores 3 independent presets with the full CC MIDI Controller map — the 18 Mixes (button+knob) and the 3 extra Knobs (4 banks). Unlike the Save Presets menu (which stores the pads' general configuration), this one stores only the CC control values — ideal for quickly switching between different mixes in your VST/DAW.

TECHNICAL SPECIFICATIONS

Hardware
and circuit

Microcontroller	**Arduino Mega 2560**	Main
Analog pads	**27mm or 35mm ceramic piezo — 1MΩ resistor + 5V1 Zener**	

Hi-Hat sensor	**TCRT5000 — port A0 — reading 0 to 1023 — InvSensor: Normal (manual assembly with resistor) or Inverted (module with LM393 — e.g. TCRT5000 Arduino LM393 5V)**	Auto-calibrates
Digital/aux pads	**Piezos on pins D33, D35, D37, D39, D41, D43, D45 — INPUT_PULLUP — 15ms debounce**	
Cymbal chokes	**Pins D47, D49, D51, D53 — direct metal contact, no external components**	
MIDI Out	**USB (native) + Serial TX1 — simultaneous**	
Display	**16×2 LCD with automatic backlight-off after 30s of inactivity**	
Storage	**Internal EEPROM — 3 presets + settings + all pad notes**	
Expansion	**Pro Micro (+4 pads) or Leonardo (+6 pads) via SysEx**	
Indicator LED	**Pin 22 + 220Ω resistor — blinks on every MIDI note generated**	
License	**MIT License — open-source, free to use and modify**	Open Source

SOURCE CODE & MANUAL

Download the complete project for free

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J-DRUMS V12 — Complete Package

The download includes the **full source code** for the Arduino Mega 2560 and the **interactive Illustrated HTML Manual** — with every section, circuit diagrams, explained parameters, and configuration examples.

⬇ J-DRUMS V12 Code Only

Arduino firmware, ready to upload

🎛 MIDI System — How it works

📤 Sends MIDI to PC / DAW

J-DRUMS sends real-time MIDI signals to any drum software — Ableton, Reaper, FL Studio, **Pro Tools, ASIO / ASIO4ALL**, and all compatible VSTs (Superior Drummer, EZdrummer, Addictive Drums).

🔕 Properly implemented Note OFF

The firmware sends **Note ON** when it detects the hit and **Note OFF** right after — exactly as the MIDI standard requires. Cymbals and notes never get "stuck" in the VST.

🎚 Editable device ID

The **module's MIDI ID is editable** directly in the code — you can customize the name that appears in the operating system and DAW when the Arduino is connected via USB, tailoring it to your setup.

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Arduino Source Code

Complete J-DRUMS V12 firmware for Arduino Mega 2560. Ready to upload with the Arduino IDE.

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Interactive HTML Manual

Complete illustrated manual in HTML — opens in any browser. Includes circuit diagrams, every parameter explained, examples, and password-protected editable notes.

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Circuit Diagrams

Wiring schematics for the analog pads, TCRT5000 sensor, digital pads, chokes, buttons, encoder, and LEDs.

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